Chat handoff: India regional intelligence map

Session date: 4 November 2026. Working name: kerala-live-map (a placeholder; the final name is not chosen). Status: research and planning finished. No code or repository exists yet.

1. How this started

Joe read a StartupHub article about a forklift driver who used AI coding tools to build a startup. The product is **KW Live** (https://kw-live.com/), a private-beta map for Waterloo Region, Ontario, built alone by Mike Nickason. The article misspells his name as "Nickerson". Joe wants to build a similar project for India, but as a data analysis + data science + ML portfolio project, not only a web map.

2. Decisions made

Decision	Detail
Build order	Karnataka first, then Kerala, then Delhi
Project type	Data engineering + analysis + ML + LLM, shown on a live map
Cost	Rs 0. Free tiers only. The Claude API is paid, so the Ask feature must use a free LLM tier
Features	As many as possible, including features KW Live does not have
Name and design	Must not reuse the KW Live name, visual identity or site text (their terms forbid it)
Planning split	ChatGPT writes the full plan as a PDF. Claude Code does the build

3. Key findings

- No Indian state can copy KW Live fully. India has no open feeds for live road closures, live police calls, public traffic cameras or radio scanners.
- Karnataka has the most data: KSNDMC rain gauges in every gram panchayat, KWRIS reservoirs, Bengaluru open data, and BMTC and Namma Metro timetables.
- Kerala is strong on floods, dams and prices: KSEB daily dam levels (16 dams), 2018 flood labels, landslide inventories, and Joe's kerala-veg-prices data (26.5M Agmarknet records).
- Delhi is closest to KW Live: live bus GPS from Open Transit Data (needs registration) and a dense network of air-quality stations.
- These sources need no key and work in all 36 states and UTs (tested live): Open-Meteo weather, historical rain since 1940, GloFAS river flow, air quality, the 40-member forecast ensemble and CMIP6 climate projections; NDMA Sachet alerts (28 of 36 states have their own feed); CelesTrak; NASA GIBS satellite tiles; AWS terrain tiles; WorldPop; geoBoundaries (736 districts); Geofabrik OSM; OpenFreeMap tiles; Google Open Buildings.
- These need a free key: data.gov.in, NASA FIRMS, aisstream, Delhi OTD.
- These failed in testing: India-WRIS (SSL error in Python, so retry in a browser), Kochi Metro open data (timeout), Kerala RERA (503), NCRB (connection failed), CGWB groundwater (failed), the ASDMA flood report page (404), the Bihar FMISC portal (site gone), and GDELT (rate-limited).

4. Where everything is saved

Item	Location
Full research notes (sections 1 to 11)	Obsidian vault: `Projects/kerala-live-map.md` (main copy)
Same notes, session copy	`KW-Live-research.md` in the session folder (deleted with the session)
Claude memory	`kerala-live-map-project.md` in this project's memory folder
Work log entry	Obsidian `Claude/Work log.md`, 4 Nov 2026
Home dashboard row	Obsidian `Home.md`
These 3 PDFs	Obsidian `Projects/kerala-live-map-files/`

Item	Location
Test scripts (probe.py and others)	session scratchpad (temporary, not kept)

5. Related projects in the vault

- `Projects/kerala-veg-prices`: 26.5M Agmarknet records (2021 to May 2026), Open-Meteo rain, and a 16-day price-spike model (PR-AUC 0.33). Reuse it for the rain-to-price work.
- `Projects/kerala-crime-map`: another person's repository with Kerala crime totals.
- `Learning/Compare ML evidence correctly`: Joe's own rules for comparing models.
- Known issue from 3 Nov 2026: DuckDB 2.0.0 corrupted files in the retail-analytics-lab project. Use DuckDB 1.x or a Python 3.12 virtual environment.

6. Next steps

1. Paste the prompt from PDF 3 into ChatGPT and get its planning PDF.
2. Paste that PDF into Claude Code.
3. Claude Code creates the repository, then the shared pipeline for Karnataka.
4. First data check: how many years of KWRIS reservoir history can be downloaded. The reservoir forecast depends on it.

7. Open questions

- Final project name.
- Whether a free LLM tier (Gemini, Groq or Cloudflare Workers AI) is good enough for Ask.
- India-WRIS access in a browser.
- How far back the KSEB monthly archive goes.
- Whether Delhi OTD approves the registration.